National Forensic Sciences University, Goa Campus

MSc DFIS - Semester II

Mid- Semester Examination

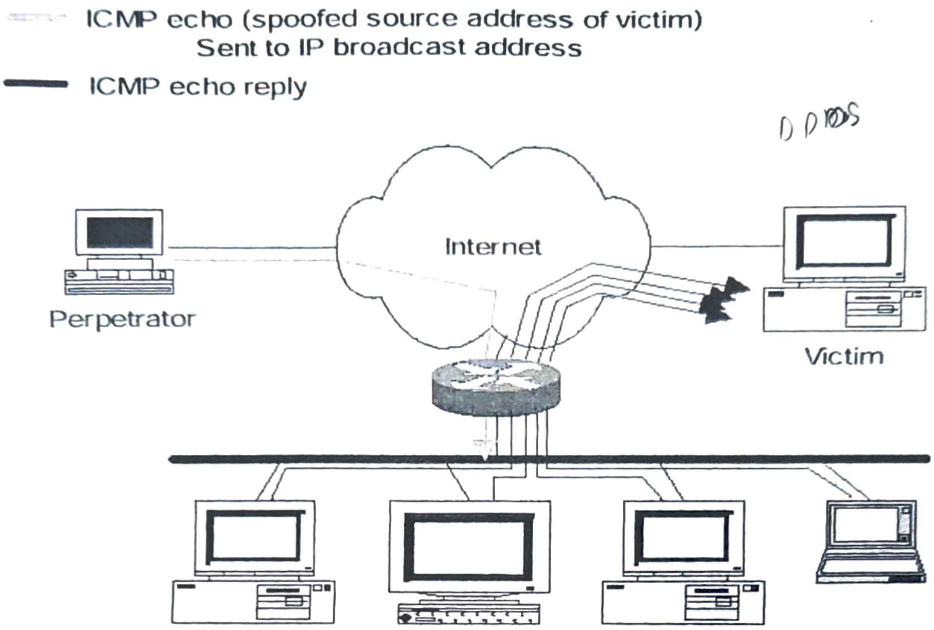
Subject Code: CTMSDFIS SII P3
Subject Name: Network Security and Forensics
Time: 90 Minutes

Date: 16/03/2025

Total Marks: 50

Instructions –

1) Answer all questions.
2) Assume suitable data.
3) A scientific calculator is allowed.
4) Parts of the question should attend the same place.

Q.1	Attempt all.	**20 marks**
(a)	Consider the scenario below:  i. Identify and explain the attack on the above scenario. ii. Also mention the countermeasures to protect against this kind of attack	5 Marks
(b)	A company wants to set up a web server accessible from the internet. Explain the role of the DNS server in this scenario.	5 Marks
(c)	Encrypt the following message using **Playfair cipher**. Message: **jacuzzi** Key: **extrajudicially**	5 Marks
(d)	(i) Calculate the power modulo, $\boldsymbol{191^{931} \bmod 103}$. (ii) What is the purpose of TTL field in IP packet? How is TTL useful for forensic purposes?	5 Marks
Q.2	Attempt three.	**15 Marks**
(a)	Differentiate Non-repudiation, Eavesdropping, and Masquerading.	5 marks
(b)	Define digital signatures and explain their role in ensuring the authenticity and integrity of digital messages. Describe the role of Public Key Infrastructure (PKI) in the context of digital signatures.	5 marks

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(c)	Explain the differences between error control and flow control. Also mention its importance in network security.	5 marks
(d)	Explain the WLAN architecture. Also mention various IEEE 802.11 standards.	5 marks
Q.3	Attempt all.	**15 Marks**
(a)	Use two global prime number ***19*** and ***23***, the value of ***e*** is ***7*** and message ***M= 5***, calculate the public key, private key, and the corresponding cipher text. Also prove that *RSA* decryption is the inverse of *RSA* encryption.	08 Marks
	OR	
	Use two global prime number **29** and ***43***, the value of ***e*** is ***17*** and message ***M= 3***, calculate the *public key, private key*, and the corresponding cipher text. Also prove that *RSA* decryption is the inverse of *RSA* encryption.	
(b)	*Alice* and *Bob* wish to swap keys by using *Diffie-Hellman* key exchange algorithm and are agreed on prime ***p = 19*** and base or generator is ***g= 5.*** Calculate the *secret key* of each user and *shared session key* for both the users. Also explain with the same question that how can *Eve* (untrusted third person) exploit *Man-in-Middle attack*.	07 Marks
	OR	
	Why is the WEP protocol considered less secure and therefore not preferred compared to the WPA protocol? Also, explain the discovery phase and authentication phase of the RSN (Robust Security Network).	

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